

CYBERSECURITY AND ETHICAL HACKING INTERNSHIP

TASK 1: FOUNDATION & ENVIRONMENT SETUP

Submitted By:

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1. Introduction

Cybersecurity is the practice of protecting systems, networks, and data from cyber threats and unauthorized access. This task focuses on building a cybersecurity lab environment and understanding the core concepts of networking, Linux, cryptography, and security tools.

2. Objectives

- Build a cybersecurity lab environment.
 - Learn Linux fundamentals.
 - Understand networking concepts.
 - Explore cryptography basics.
 - Familiarize with cybersecurity tools.
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3. Cybersecurity Basics

CIA Triad

Confidentiality

Protects information from unauthorized access.

Integrity

Ensures data remains accurate and unmodified.

Availability

Ensures systems and services remain accessible.

4. Lab Environment Setup

Host Operating System

Windows 11

Virtualization Platform

VMware Workstation

Virtual Machines

Kali Linux

Used as the attacker machine for penetration testing.

Metasploitable2

Used as the vulnerable target machine.

5. Linux Fundamentals

Commands practiced:

pwd

ls

cd

chmod

touch

apt update

Topics covered:

- File Navigation
 - Directory Management
 - File Permissions
 - Package Management
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6. Networking Basics

Topics covered:

- OSI Model
- TCP/IP Model
- DNS
- HTTP/HTTPS

- IP Addressing
- Network Communication

Networking commands:

ip a

ifconfig

ping

netstat

7. Cryptography Basics

Symmetric Encryption

Uses a single key for encryption and decryption.

Asymmetric Encryption

Uses public and private keys.

Hashing

Algorithms used:

- MD5
- SHA256

SSL/TLS

Provides secure communication over the internet.

8. Tool Familiarization

Wireshark

Used to capture and analyse network packets.

Nmap

Used to identify hosts, services, and open ports.

Burp Suite

Used to intercept and analyse web traffic.

Netcat

Used for communication and network testing.

9. Practical Activities Performed

- Configured Kali Linux and Metasploitable2.
 - Verified IP configurations.
 - Performed Linux file operations.
 - Updated packages using APT.
 - Conducted network scanning using Nmap.
 - Captured packets using Wireshark.
 - Intercepted HTTP traffic using Burp Suite.
 - Generated MD5 and SHA256 hashes.
 - Tested network communication using Netcat.
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10. Screenshots

Insert screenshots in the following order:

1. Metasploitable IP Configuration
 2. Kali Linux IP Configuration
 3. Linux File Permissions
 4. APT Package Management
 5. Ping Test
 6. Nmap Scan
 7. Cryptography Hashing
 8. Wireshark Packet Capture
 9. Burp Suite HTTP Intercept
 10. Netcat Communication
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11. Learning Outcomes

- Developed a cybersecurity lab environment.

- Learned Linux administration basics.
 - Understood networking concepts and protocols.
 - Practiced cryptographic hashing.
 - Used professional cybersecurity tools.
 - Gained practical penetration testing exposure.
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12. Conclusion

Task 1 successfully established a strong foundation in cybersecurity. The lab environment provided hands-on experience with Linux, networking, cryptography, and security tools. The practical exercises enhanced understanding of real-world cybersecurity concepts and prepared the environment for future penetration testing activities.

Thank You.